



A Modular Pill Dispenser Supporting Therapies at Home

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Abstract. Modern technologies support people's life in multiple contexts like the assistive one. The pervasiveness of the so-called "Smart Objects", related to the Internet of Things technologies, is boosting this in many ways. The support for old people to take the daily tablets through an automatic device is an example. This work presents the prototype of a modular pill dispenser customized by end users according to their specific therapy needs. The prototype is a physical and modular set of pillboxes each containing the pills to be assumed in a therapy. The presented scenario at a specific time, set by the user, one or more pillboxes blink to alert the patient that is pill time. If for a given time interval the patient do not take the pill a sound notification is activated and plays for a given duration. If still nothing happens then a notification is sent to the caregiver's smartphone. The behavior of the pill dispenser is defined by the end user and can be modified any time. One peculiarity of the pill dispenser is that the number of physical boxes are decided by the user and can change any time to best fit the specific therapy. The final goal of this work is to push not professional users, in particular older people, to take advantages of new technologies to improve their life.

Keywords: IoT · Pill dispenser · Elderly therapy

1 Introduction

The Internet of Things (IoT) technology allows devices to connect to the Internet and thus to send information or to be controlled remotely. One of the most explored field of research is how the pervasiveness of those devices can support people in their daily activities. Some solution help people manage and control the home automation [1]. Others use different techniques to push users to create collaborative rules shared solutions for conflicting needs) in the smart home environment [2]. The branch of IoT applied to the health, which is the domain of the work presented in this paper, is also gaining importance. Built-in sensors can be exploited to monitor the health status of a person [3]. Thanks to the continuous reduction of sensor size, the applications and the pervasiveness of the IoT devices is increasing. Indeed, sensors can be implanted into patients to keep track of the health parameters using a smartphone or a tablet [4].

Such technologies can also be used to support people in their daily activities like taking pills during a therapy. The idea to support people to self-administrate drugs with automatic devices to support pill-based therapies come from the 60's [5]. Nowadays, thanks to the modern communication advances, the availability of Internet connection in such devices, allows physicians to get closer to the user's needs and caregivers, family members and patients are now allowed to remotely monitor the pill therapy [6, 7]. Some researchers are addressing the development of systems to control the user/patient health status, while others are investigating on the hardware and electronic to get better and smarter devices in terms of performance when they have to recognize the symptoms of illnesses [8, 9].

In this work we have developed a prototype of a modular pill dispenser which can be configured, according to the user's needs, with multiple pill therapies and that belongs to the Web of Things research area [10]. Each pill is associated with a small smart box. The therapy, as well as the alerts are set using a mobile phone app.

The paper is structured as follows: Sect. 2 presents a short overview of some solutions available in the market. Section 3 describes the pill dispenser prototype, while in Sect. 4 a discussion of the results of a pilot test performed with real users is provided. Finally, in Sect. 5, the conclusions of the work is proposed.

2 Existing Proposals

The search for existing solutions was made according to three perspectives: *Integrated Systems*, which allow the users to set therapies using physical devices; *Management Apps*, which support therapies only on mobile devices, and *Consultation Apps* that connect to remote repositories to retrieve information about drugs. In the following we report two representatives for each perspective.

Integrated Systems. *Smart Pill Box*¹ is equipped with six, removable, boxes, having different colors and sizes. The pill box does not contain any electronic inside. It performs the simple function of a pocket box for the patient's pills. The system's logic is managed by the *Pill buddy application* on mobile devices. The application's Graphical User Interface (GUI) recalls the topology of the pill box, using the same colors of the pill box compartments in order to easily identify the different pills. The app is useful as a reminder but lacks the history of the pills taken. *Pillbox by Tricella*² is equipped with Bluetooth connectivity and sensors that detect if boxes are opened or closed. Pills are managed on a mobile device. By default, the pill box has seven different boxes, but it is possible to configure an unlimited but fixed number of pill slots per user. The app shows the history of pills taken in a monthly calendar and displays the next pill to take for a specific family member.

¹ <http://shop.vitality.com/en/smart-pill-box-zwart>, last visited May 2018.

² <http://www.tricella.com/>, last visited May 2018.

Management apps. *Medisafe*³ is a medical care management and a pill reminder application that keeps track of blood pressure, glucose and other values, manage physicians and their appointments. Such data must be manually provided by the user. *MyTherapy*⁴, is used as the pharmacological and care treatment. It sends the user one or more reminder and keep track of all the medication history.

Consultation apps. *Banca Dati Farmaci*⁵, created by the Italian Medicines Agency (AIFA), is the official Italian database that provides summaries characteristics and up-to-date illustrative sheets of drugs sold in Italy. It provides the users the search for a medication, all its variants and package leaflets. An alternative is *Farmaci Lite*⁶ that, in addition to the usual data, such as name, active ingredient and manufacturer, provides more detailed information for each drug. On purpose we reduced the scope to Italian databases because we interacted with Italian users. In case of an extension of the user group to other countries further drug databases can be considered.

From the study of the apps we selected the following functionalities that a pill dispenser must have.

- Therapy Management: the possibility to manage the pill therapy (e.g.: adding or removing a pill taking schedule) using the devices or through a mobile app;
- History: lists therapies done and/or the pills taken in the past;
- Caregiver: a person, different from the user, that can have access to the therapy management (including the possibility to edit the schedule and the drug to take);
- Reorder and Refill management: the management of the quantity of pills in the boxes and their refill;
- Drug leaflet: the basic information of pills;
- Drugs details: more details about the implication of drug consequences and uses;
- Drug database: access to publicly available drug databases;
- Report: reports about the pill therapy;
- Freeware: the app is free to use; and
- UI guidelines: standards for the user interface.

Table 1 briefly reports the features available in the selected systems; the tick indicates the presence of the specific functionality. According to the first four rows, the Consultation apps do not have interaction with the user, because such apps are intended just for gathering information and not for managing therapies. All the apps, but *Smart Pill Box* have a drug database. However, only *Farmaci Lite* provides more details; in addition, only the consultation apps provide the drug leaflet. Four out of six apps are freeware and four apps comply with User Interface guidelines. Only *Management apps* show the user a report about therapies.

³ <https://medisafe.com/>, last visited May 2018.

⁴ <https://www.mytherapyapp.com/>, last visited May 2018.

⁵ <https://farmaci.agenziafarmaco.gov.it/bancadatifarmaci/>, last visited May 2018.

⁶ <http://www.ilprontuariofarmaceutico.it/>, last visited May 2018.

Table 1. Comparison of mobile Apps and IoT solutions to support a user on the pill therapy.

Functionalities	Integrated systems		Management apps		Consultation apps	
	Smart pill box	Pillbox by Tricella	Medisafe	MyTherapy	Banca Dati Farmaci	Farmaci lite
Therapy manager	✓	✓	✓	✓	×	×
History	×	✓	✓	✓	×	×
Caregiver	×	✓	✓	✓	×	×
Reorder & refill management	×	✓	✓	✓	×	×
Drugs leaflet	×	×	×	×	✓	✓
Drugs details	×	×	×	×	×	✓
Drugs database	×	✓	✓	✓	✓	✓
Report	×	×	✓	✓	×	×
Freeware	✓	✓	×	✓	✓	×
UI guidelines	×	✓	✓	✓	×	✓

The study of the existing proposals led to some requirement that the modular pill dispenser should satisfy.

- (R1) the need for a modular system. Most of the commercial solution provides a fixed number of *boxes* that are intended to support therapies that do not change over time and are based on the same amount of pills. Since therapies may change over time the modularity must be taken into account, both software and physical modularity.
- (R2) feedback about the pills intake. The caregiver must be sure that the patient is taking the pills correctly. A few existing systems address alerts if the pill has not been taken or provides the user with the history of pill taking.
- (R3) app customization. Any or very few systems address the dynamic customization of the therapy. If the drugs do not affect the properly patient, the doctor may chose to change it. Often this means deleting the current therapy and set a new one.

The literature proposes many solutions to the problems related to the pill assumption. Morales has performed a study to evaluate how the pillboxes can improve the therapy assumption [11]. In her work Morales demonstrated that it is clear that automatic and electronic pillboxes can improve the therapy assumption. Hayes proposes many solution to constantly monitor whether individuals take the right pill at the right time [12]. It involves constant data collection and elaboration in order to check that the patient has taken the right pill.

Recently, Mira, proposed a pillbox app to monitor elderly patients taking multiple medications [13]. According to their studies, even people with low or no

experience are able to use and understand mobile application alerts and respond to them.

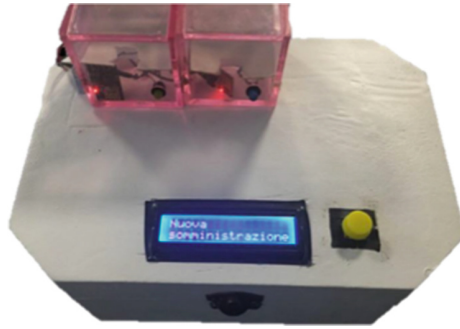


Fig. 1. The pill dispenser with two pillboxes (in pink) on top of it (Color figure online)

3 The Proposed Solution

According to the analysis of existing work, we have developed a modular pill dispenser to support people in taking pills therapy. This is an improvement of our previous work, where the user received an alert when the pill had to be taken [14]. The main differences are visible both, in the mobile application and in the modularity of the pill dispenser which also allows multiple therapies at once. The modularity of the device addresses the requirement (R1) by enabling the user to manage any number of needed pill types, according to the specific needs. (R2) requirement is met by adding the *history* to the mobile app and a “notification” feature in case the pill has not been taken at the right time. Also the requirement (R3) is addressed, through the use of the End-User Development approach, which is well known in the Human-Computer Interaction field, and which “encompasses methods, techniques, methodologies, situations, and socio-technical environments that allow end users to act as professionals in those domains in which they are not professionals” [15]. According to such approach the users can tailor the program to the real and current needs, thanks to the efforts of developers that give them more “power”. As a matter of facts, more and more projects that allow users to create custom electronic solutions⁷ are coming out. The natural evolution of this concept is the “meta-design approach”, which empower the users to edit and manipulate the software artifact according to their needs [16].

The proposed solution has the following components: the physical pill dispenser, composed of a microcontroller, sensors, communication modules and a number of boxes that contain the pills. To the user, the pill dispenser appears as

⁷ <http://littlebits.cc/>, last visited May 2018.

a set of boxes: the biggest one is the base box that hosts a set of pill boxes, which are the actual pill containers (see Fig. 1); the app allows the user to configure and manage the pill therapy; a server monitors the pill dispenser status and sends all changes performed through the app to the pill dispenser. The server can be connected to external services, such as hospitals, doctors, pharmacies, that can prepare new therapies on behalf of the caregiver. This case, however, is out of the scope of this paper.

One of the problems connecting physical devices to software systems is their initial setup. We introduced a “smart” procedure to allow the communication between the pill dispenser, the pill boxes and the smartphone. By default, the pill dispenser generates a Wi-Fi SSID that the user can use to connect the smartphone to the pill dispenser. Using the configuration menu of the app, the user can setup the pill dispenser and connect to an existent Wi-Fi network. An alternative for the configuration might be to put such information into an internal SD card, but this means that the user must have some technical skills. After the pill dispenser is connected to the Internet it sends his MAC address and since then this will be the unique identifier for the pill dispenser. Once the connection is set and the device ID is stored into the server the app asks the caregiver to insert the name of the patient and a password. Then the caregiver associates a therapy and a pill dispenser to the user. This is all the needed steps to setup a new pill dispenser.

In order to correctly setup the therapy, each pill type must be associated to a pill box and must be specified the number of pills in the pill box (see Fig. 2).

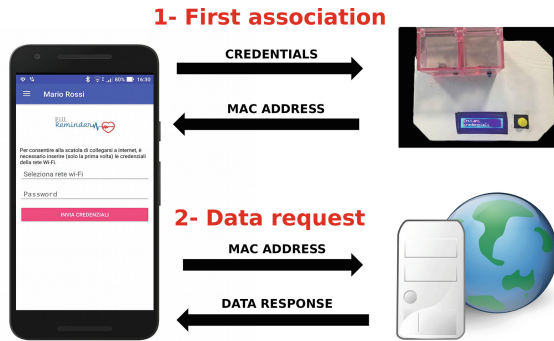
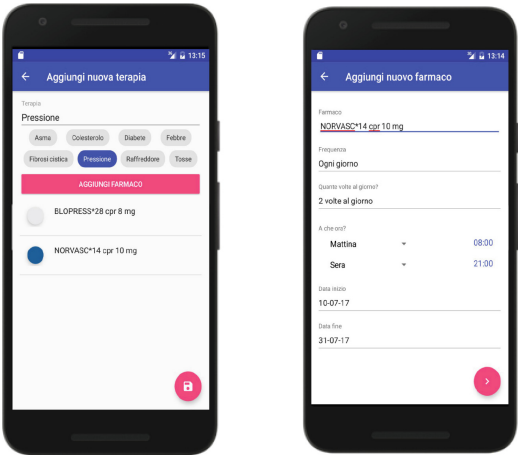


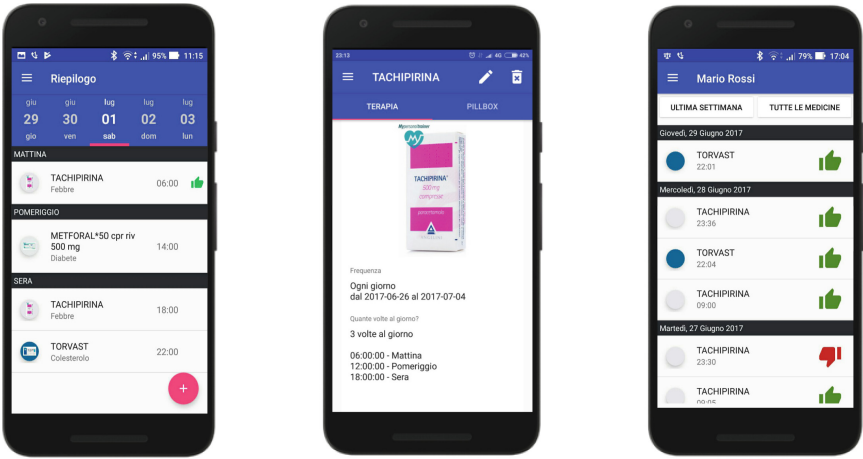
Fig. 2. Identification of a pill dispenser

The app shows a list of “free” boxes, according to the number of pill boxes available on the base box. The user must associate the pill box to the pills and then specify the timing of the LED, the delay before an acoustic sound will be triggered and the duration of the sound, if the default options do not satisfy the user’s needs. Through the app, the user can create and review previous therapies and some statistics (see Fig. 3a). One or more drug can be associated



(a) (b)

Fig. 3. Screenshots of the app representing the creation of a new therapy (a) and the addition of a new drug to the therapy (b)



(a) (b) (c)

Fig. 4. Screenshots of the app revealing a few features: a calendar-like view to schedule user therapy (a); detailed information about the drug (b); quick view of pill taking (c)

to the therapy (see Fig. 3b). In order to support users to create an event of pill intake, a calendar-like view has been implemented (see Fig. 4a. Details about specific drugs the patient must assume are provided in a specific screen of the app (see Fig. 4b). A summary of “taken” and “lost” pills (see Fig. 4c) is also provided.

Therapies require the following data:

- The drug’s name, according to a public agency, we used the “Agenzia Italiana Farmaco” (AIFA);
- Frequency: the time between two consecutive pill intakes of the same drug;
- Interval: for intakes that are alternate with a given time interval (e.g. alternate days);
- Days of week, if the drug must be taken at specific days of the week;
- Start and End date of the therapy; and
- Times per day, if the drug must be taken once or more per day.

In order to get information about the pills, the base box requests and upgrades the system status to a web server that stores all the needed information to setup the pill therapy and user alerts. The diagram of the system is described in Fig. 5. The prototype is made by two hardware parts: the set of pill boxes, which contain the pills, and the base box that contains the micro controller board, a display, a button and a buzzer. By default, each box can contains only one pill type.

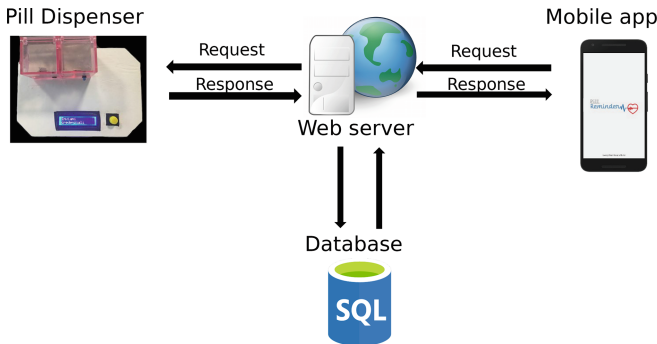


Fig. 5. The diagram of the system components and communications

An example of pill box is represented in Fig. 6. Each pill box is composed of a magnet and two metal plates for each opposite sides and one LED. The magnet allows one or more boxes to be stucked together. The first box attached to the system is connected directly with the base box. The two metal plates, positioned on the same sides of the magnets, power the smart box. The top of the block can be opened to allow the user/patient to put or take one or more pills. A LED, positioned on one side, indicates that an interaction with the box is required.

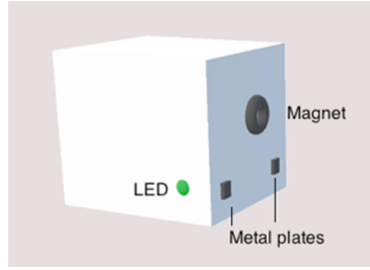


Fig. 6. The 3D model of a pill box

The LED turns “ON” when is pill time; it turns “OFF” when the user takes the pill. In order to tell the pill box that the pill has been taken, the user presses the button. The communication between each pill box and the base is done through an ESP8266-01 programmable module. This component not only allows the box to be wirelessly connected to the pill dispenser, but also to uniquely identify each box. Indeed, every box ID must be recognized by the system in order to “see” that a new pill box exists. The logic of the base box is delegated to an Arduino board. We were comfortable with an Arduino Mega board but with the right optimizations also an Arduino Uno works. Other parts like a buzzer, a button, a display and LEDs are used to interact with the user. The system connects to a wireless network (using the 802.11 b/g/n protocol) with or without encryption. The hardware parts works as a middleware between the software and the pill boxes [17]. When is pill time, the related LED on the base box turns on. If the user does not take the pill, after the amount of time specified by the user, the buzzer starts vibrating, trying to attract the user/patient’s attention. The user that takes the pill pushes the button on the base box in order to ask the system to stop other alerts scheduled for the specific pill (see Fig. 1).

4 Discussions

We have proposed a modular pill dispenser to support people that must follow pill-based therapy. The system is composed of three main components: a web server that stores the information about the therapies and send alerts when needed; the mobile application that manages therapies and configure the pill dispenser (including each pillbox); the pill dispenser that is composed of a set of pill boxes and the base box that contains the logic managing sensors and actuators for the interaction with the user. The customization of the pill therapy and the behaviour of the system (through a mobile app) allows the user to tailor the system to the own need and therapy. Moreover, we introduced the possibility to physically change the number of pill boxes that contain the pills of the therapy. This saves space and do not have empty boxes or pills outside the pill dispenser.

In order to check the validity of the approach we performed a formative evaluation with a pilot set of Italian users. The reason why we have decided

to use only Italian reference drugs is because all the users who performed the usability test were italians. The average age was 37. We asked 5 people to perform the following 7 tasks:

1. Add the Zimox drug to take twice a day, for a total of 10 days, after the meals;
2. Show which drugs must be taken on the 18 of July and then show the current date;
3. Show the details of the therapy at the point 1;
4. Edit the therapy at the point 1, changing the pill therapy from two times to three times per day;
5. Show the informative sheet of the Zimox drug;
6. Add a family member as caregiver; and
7. Show which drugs have been taken correctly on the 17th of July.

Table 2. The table with the times and the completing rate for each task. S = Success, P = Partial success, F = Failure

	Task 1		Task 2		Task 3		Task 4		Task 5		Task 6		Task 7	
User 1	S	2:30	P	0:50	S	0:50	S	0:32	S	0:26	F	4:30	S	1:32
User 2	S	1:44	P	0:10	S	0:25	S	0:17	S	0:25	F	1:44	S	2:03
User 3	S	2:04	P	0:16	S	0:32	S	0:39	S	0:18	F	2:09	S	2:14
User 4	S	3:20	P	0:21	S	0:16	S	0:35	S	0:10	S	0:57	S	3:35
User 5	S	2:26	P	0:18	S	0:16	S	0:38	S	0:16	S	1:33	S	1:58
Average		2:25		0:23		0:28		0:32		0:19		2:23		2:17

In Table2 the times taken to complete the tasks are reported. For each user (row), the completing result and the relative time spent to complete it has been shown. The outcome for each task can be “Success” (S) if the user has reached the end of the task; “Partial Success” (P) if the user has partially completed the task; “Failure” (F) if the user has not completed at all the task. The most difficult task was the 6, which required to add a new person in the mobile application in order to set the role of caregiver. Three users out of five failed it due to the difficulty to find the correct menu entry to complete the task. The average success rate of the usability test is about the 84%. Despite this is a really small number and tasks are not reported in detail, the formative evaluation revealed that the most common tasks were easy to reach. The most problematic tasks are those related to the schedule. If they are performed by a trained person this is an acceptable compromise; we are thinking of ways to overcome it.

5 Conclusions

The proposed system contains several strengths but also presents some weaknesses. Currently, the pill dispenser trusts the patient that claim to have taken

the pill by pressing a button. A more sophisticated way to be sure the patient actually takes the pill is needed, but this is not easy to achieve, since it would need also more sophisticated equipment at the patient's house. A possible improvement can be to add a sensor that notifies the app when the box has been opened, but it does not completely solve the problem. Currently the pill dispenser is powered by a common phone charger that provides a current of 5 V at 1 A and the base box can easily connect 20 pill boxes. If more pill boxes are needed the pill dispenser could be provided with a specific power supply that would provide more power. Even though the pill dispenser can be easily powered by a portable power bank, it is limited by the wireless connectivity: once the signal is lost, it would not be able to communicate with the server (and the mobile app consequently) anymore. If the user is away from home, and the pill is not taken according to the therapy, the alerting procedure (as described in the Sect. 3) starts. Finally, the current implementation of the system considers just one pill dispenser per user. In the future we plan to address more users per pill dispenser and to improve the entire system to be more fault tolerant in different conditions. Moreover, the mobile app might check the user's position (through the internal GPS sensor) and consequently send an alert if he/she is moving away when the pill therapy time is approaching.

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References

1. Kelly, S.D.T., Suryadevara, N.K., Mukhopadhyay, S.C.: Towards the implementation of IoT for environmental condition monitoring in homes. *IEEE Sens. J.* **13**(10), 3846–3853 (2013)
2. Caivano, D., Cassano, F., Fogli, D., Lanzilotti, R., Piccinno, A.: We@Home: a gamified application for collaboratively managing a smart home. In: De Paz, J.F., Julián, V., Villarrubia, G., Marreiros, G., Novais, P. (eds.) *ISAmI 2017. AISC*, vol. 615, pp. 79–86. Springer, Cham (2017). https://doi.org/10.1007/978-3-319-61118-1_11
3. Istepanian, R.S.H., Sungoor, A., Faisal, A., Philip, N.: Internet of m-health things 'm-iot' (2011)
4. Park, S.W., Ko, H.J., Park, Y.I., Yoo, S.J., Han, Y.C.: Controller, and patch type automated external defibrillator for controlling defibrillation using the same. US Patent 9,757,580, 12 September 2017
5. Moulding, T.: Proposal for a time-recording pill dispenser as a method for studying and supervising the self-administration of drugs. *Am. Rev. Respir. Dis.* **85**(5), 754–757 (1962)
6. Morak, J., Schwarz, M., Hayn, D., Schreier, G.: Feasibility of mhealth and near field communication technology based medication adherence monitoring. In: 2012 Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), pp. 272–275. IEEE (2012)

7. Caivano, D., Fogli, D., Lanzilotti, R., Piccinno, A., Cassano, F.: Supporting end users to control their smart home: design implications from a literature review and an empirical investigation. *J. Syst. Softw.* **144**, 295–313 (2018)
8. Dimauro, G., Caivano, D., Girardi, F., Ciccone, M.M.: The patient centered electronic multimedia health fascicle-EMHF. In: 2014 IEEE Workshop on Biometric Measurements and Systems for Security and Medical Applications (BIOMS) Proceedings, pp. 61–66. IEEE (2014)
9. Yeole, A.S., Kalbande, D.R.: Use of internet of things (IoT) in healthcare: a survey. In: Proceedings of the ACM Symposium on Women in Research 2016, pp. 71–76. ACM (2016)
10. Guinard, D., Trifa, V., Mattern, F., Wilde, E.: From the internet of things to the web of things: resource-oriented architecture and best practices. In: Uckelmann, D., Harrison, M., Michahelles, F. (eds.) *Architecting the Internet of Things*, pp. 97–129. Springer, Heidelberg (2011). https://doi.org/10.1007/978-3-642-19157-2_5
11. Morales, M.T.S.V.: Study on the use of a smart pillbox to improve treatment compliance. *Aten. primaria* **41**(4), 185–191 (2009)
12. Hayes, T.L., Hunt, J.M., Adami, A., Kaye, J.A.: An electronic pillbox for continuous monitoring of medication adherence. In: 28th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBS 2006, pp. 6400–6403. IEEE (2006)
13. Mira, J.J., et al.: A spanish pillbox app for elderly patients taking multiple medications randomized controlled trial. *J. Med. Internet Res.* **16**(4) (2014)
14. Buono, P., Cassano, F., Legretto, A., Piccinno, A.: A homemade pill dispenser prototype supporting elderly. In: Garrigós, I., Wimmer, M. (eds.) *ICWE 2017. LNCS*, vol. 10544, pp. 120–124. Springer, Cham (2018). https://doi.org/10.1007/978-3-319-74433-9_10
15. Fischer, G., Fogli, D., Piccinno, A.: Revisiting and broadening the meta-design framework for end-user development. In: Paternò, F., Wulf, V. (eds.) *New Perspectives in End-User Development*, pp. 61–97. Springer, Cham (2017). https://doi.org/10.1007/978-3-319-60291-2_4
16. Costabile, M.F., Fogli, D., Mussio, P., Piccinno, A.: A meta-design approach to end-user development. In: *IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, vol. 2005, pp. 308–310. IEEE Computer Society (2005)
17. Bernstein, P.A.: Middleware: a model for distributed system services. *Commun. ACM* **39**(2), 86–98 (1996)